

# EMCH 260

# **SOLID MECHANICS**

**Ming Hu**

Associate Professor  
Department of Mechanical Engineering  
University of South Carolina  
E-Mail: [hu@sc.edu](mailto:hu@sc.edu)

# Stress

# 1

## Chapter 1.2

- ✓ Introduction to Mechanics of Solids
- ✓ Different loads and support reactions
- ✓ Know how to write equations of equilibrium and solve the resultant internal loading by section method

# WHAT IS MECHANICS OF SOLIDS?

Solid mechanics is the branch of continuum mechanics that studies the behavior of solid materials, especially their motion and **deformation** under the action of forces, temperature changes, phase changes, and other external or internal agents.

*(from Wikipedia)*

## Typical Course Contents:

- Stress
- Deformation (Strain)
- Compatibility
- Infinitesimal Strain **Elasticity** (linear)
- **Plasticity**
- Bending / Torsion / Shearing / Buckling, etc.
- Hooke's Law
- Material Failure Theory
- Fracture Mechanics
- Contact Mechanics
- .....

# WHAT IS MECHANICS OF SOLIDS? (Cont.)



In order to design the members of this building frame, it is first necessary to find the internal loadings at various points along their length.

# WHAT IS MECHANICS OF SOLIDS? (Cont.)



The weight of this sign and the wind loadings acting on it will cause normal and shear forces and bending and torsional moments in the supporting column.



# STATICS VS. MECHANICS OF SOLIDS



The bolts used for the connections of this steel framework are subject to stress.

## STATICS

Will tell you how much forces are applied to the connections or fasteners

## MECHANICS

Will tell you how much forces are resulted inside the connections

# STATICS

Statics is the branch of mechanics that is concerned with the analysis of loads (force and torque, or "moment") acting on physical systems that **do not experience an acceleration** ( $a = 0$ ), but rather, are in **static equilibrium** with their environment.

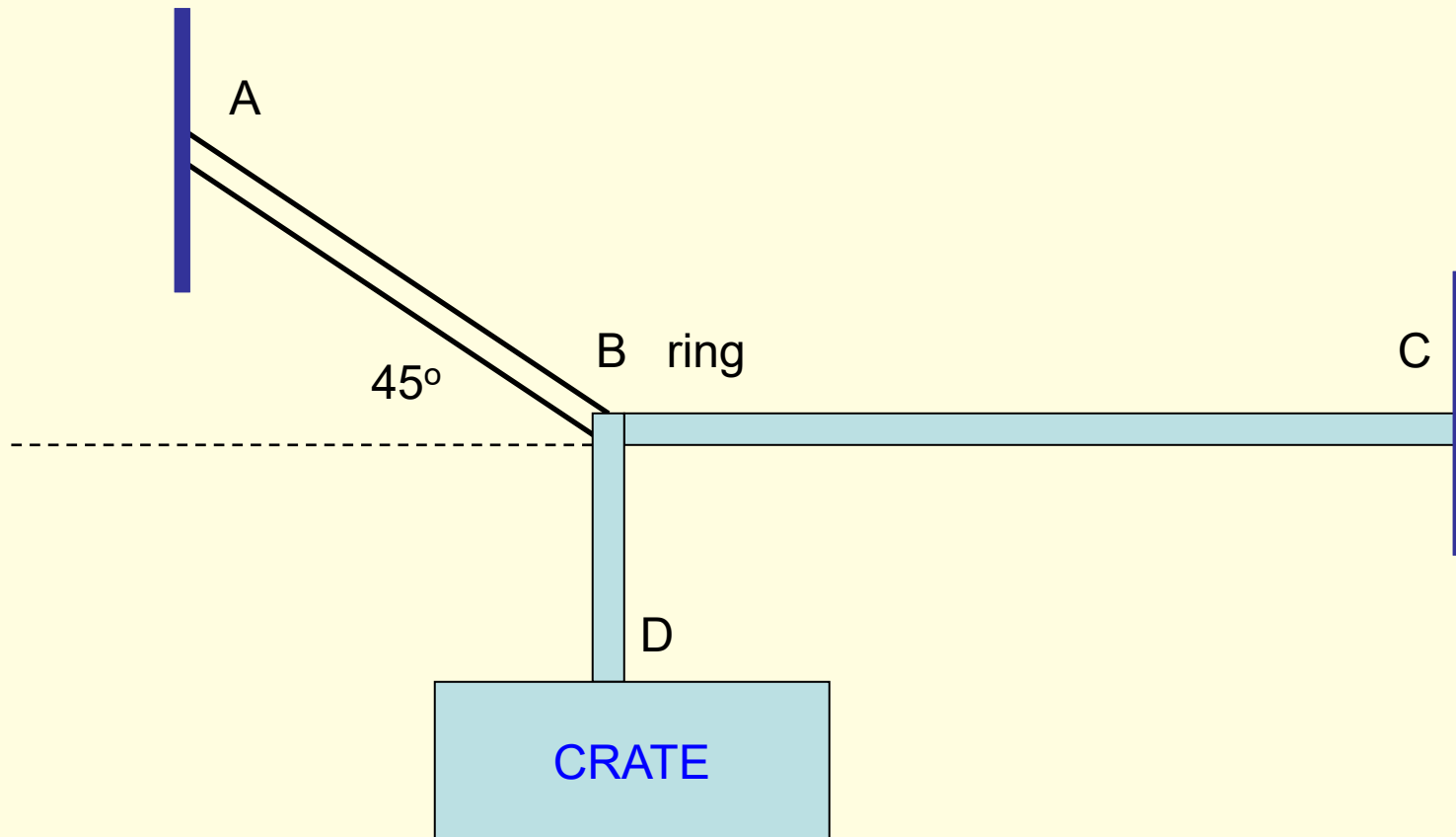
*(from Wikipedia)*

## Typical Course Contents:

- Forces in a plane, Forces in space
- Statics of **Rigid Bodies**
- Equilibrium of Rigid Bodies (2 and 3 Dimensions)
- Moments of Inertia of Areas and Masses
- Analysis of Structures (Trusses, Frames and Machines)
- Friction
- .....
- .....

# EXAMPLE OF A STATICS PROBLEM

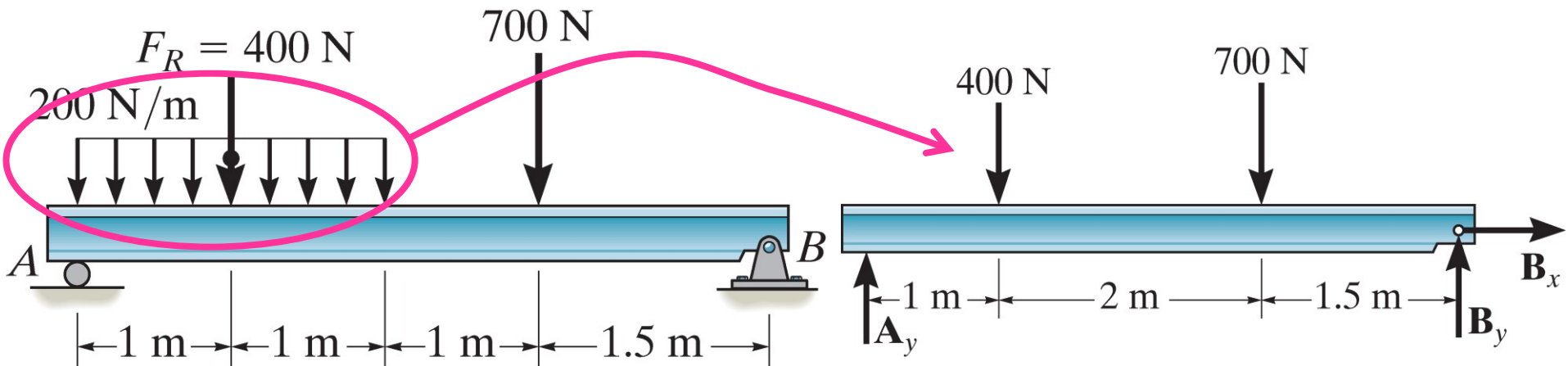
The crate below has a weight of 50 kg. Draw a free body diagram of the crate, the cord BD and the ring at B.





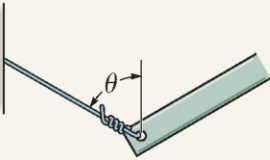
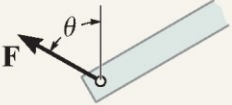
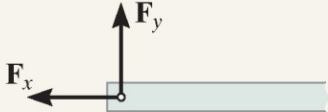
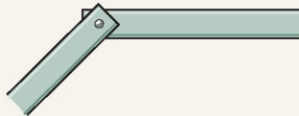
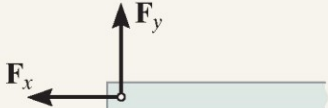
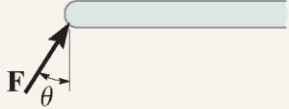
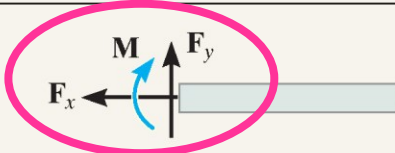
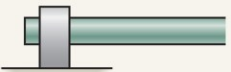
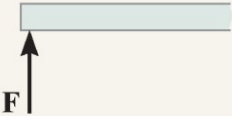
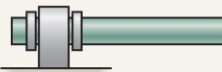
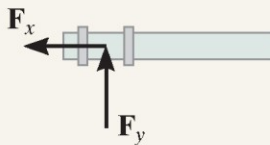
# LOADS

- Distributed loadings
- Concentrated surface loadings
- Body forces (gravity, electric / magnetic field, .....)
- Equivalent effect of distributed loadings to concentrated force  
*magnitude = area under the load diagram*  
*location passes through the centroid of loading area*



# SUPPORT REACTIONS

- Preventing a translation needs a force in the same direction
- Preventing a rotation needs a couple moment

| Type of connection                                                                                         | Reaction                                                                                                               | Type of connection                                                                                          | Reaction                                                                                                                             |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Cable</p>             |  <p>One unknown: <math>F</math></p>   |  <p>External pin</p>      |  <p>Two unknowns: <math>F_x, F_y</math></p>       |
|  <p>Roller</p>            |  <p>One unknown: <math>F</math></p>   |  <p>Internal pin</p>      |  <p>Two unknowns: <math>F_x, F_y</math></p>       |
|  <p>Smooth support</p>   |  <p>One unknown: <math>F</math></p> |  <p>Fixed support</p>   |  <p>Three unknowns: <math>F_x, F_y, M</math></p> |
|  <p>Journal bearing</p> |  <p>One unknown: <math>F</math></p> |  <p>Thrust bearing</p> |  <p>Two unknowns: <math>F_x, F_y</math></p>     |

# EQUATION OF EQUILIBRIUM

- Balance of forces (total form)

$$\sum \mathbf{F} = \mathbf{0}$$

- Balance of moments (total form)

$$\sum \mathbf{M}_O = \mathbf{0}$$

- Balance of forces (component form)

$$\sum F_x = 0$$

$$\sum F_y = 0$$

$$\sum F_z = 0$$

- Balance of moments (component form)

$$\sum M_x = 0$$

$$\sum M_y = 0$$

$$\sum M_z = 0$$

# COPLANNAR FORCES (xy plane)

- Balance of forces (component form)

$$\sum F_x = 0$$

$$\sum F_y = 0$$

- Balance of moments (component form)

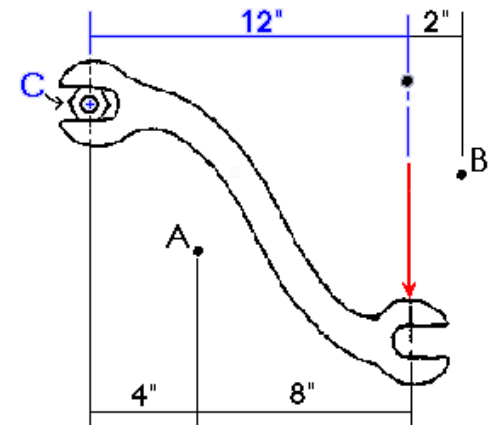
$$\sum M_z = 0$$

WHY only z-moment?

**Moment Vector Calculation (for 2D or 3D)**

$$\vec{M}_o = \vec{r}_{oF} \times \vec{F} \quad \text{Perpendicular distance!}$$

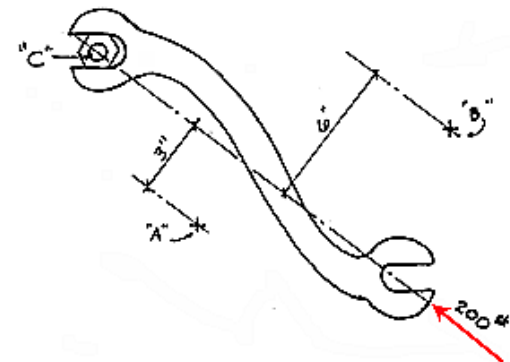
$$|\vec{M}_o| = (\text{Force}) \cdot (\text{Perpendicular distance})$$



Moment Arm (d) = 12 inches

Magnitude (F) = 100 lbs

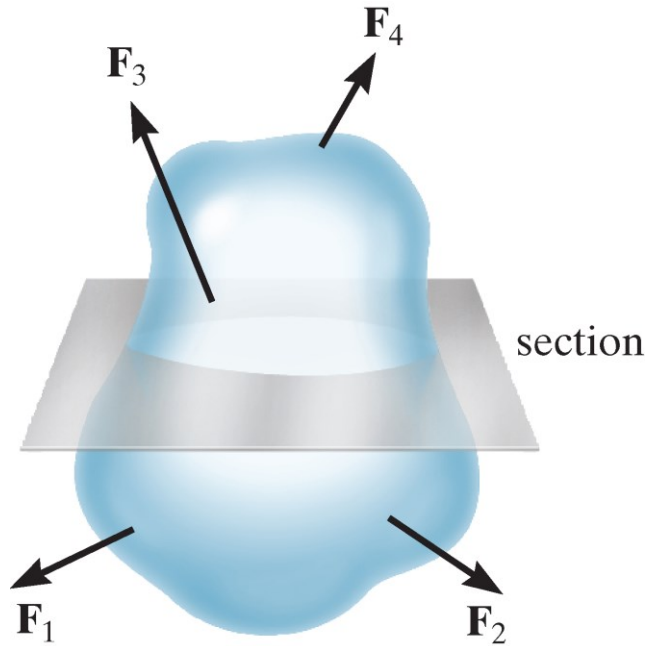
**Moment = M = 1200 in-lbs**



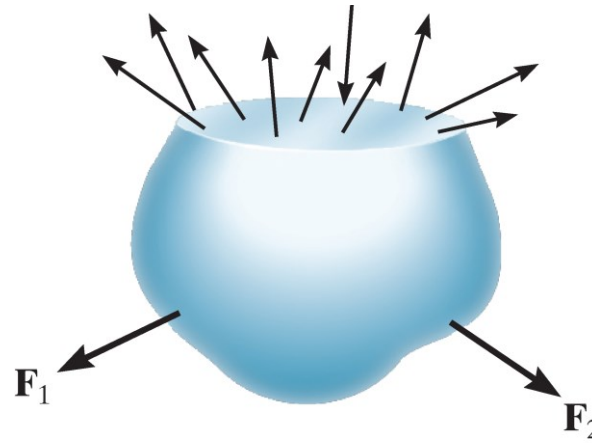
$$M = F \times d = 200 \text{ lbs} \times 0 \text{ in} = 0 \text{ in-lbs}$$

# INTERNAL RESULTANT LOADINGS

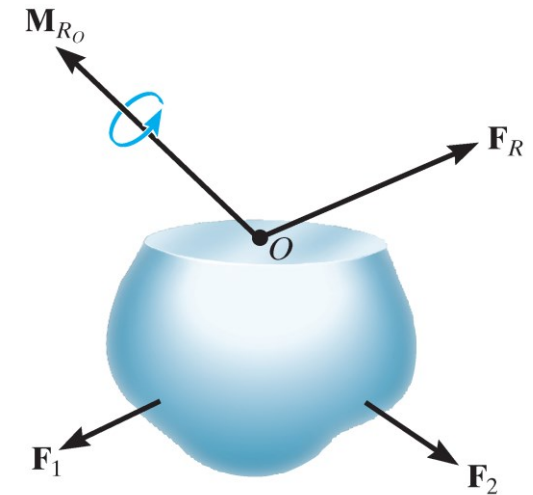
- Method of Sections



(a)



(b)



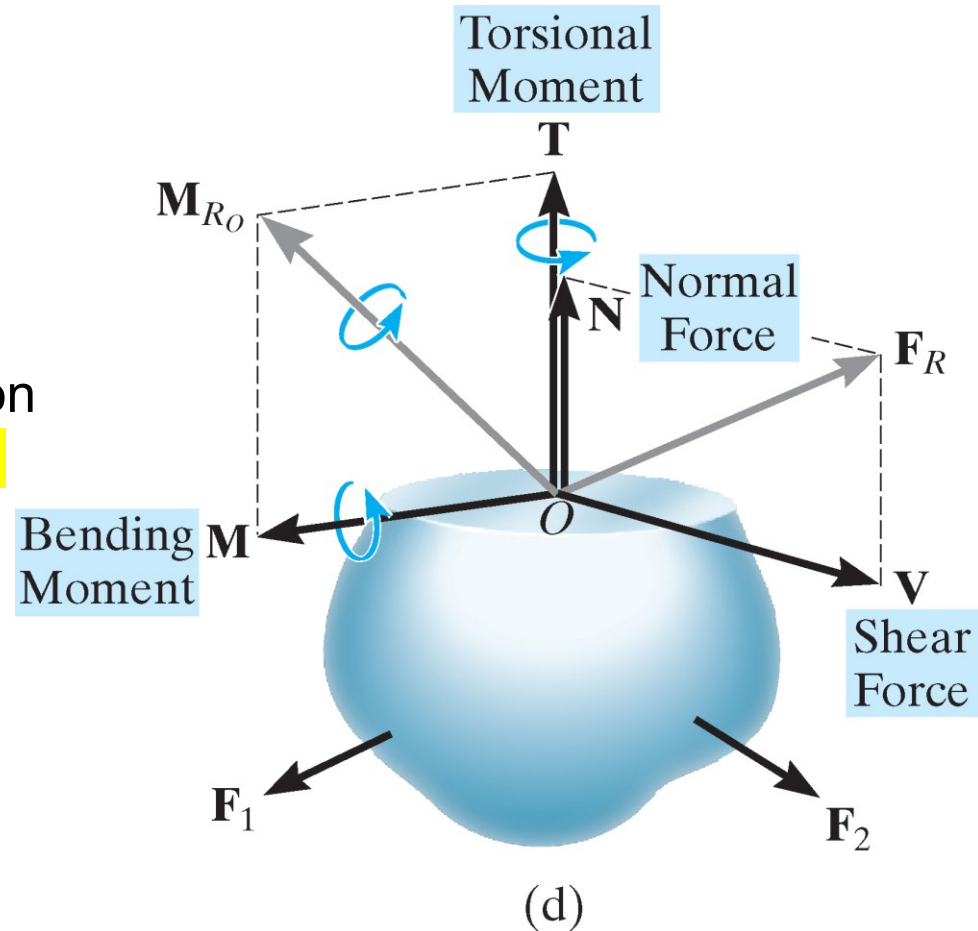
(c)

- Imaginary* section or “cut” through the region where the internal loadings are to be determined, and apply free body diagram to one of two parts (*either part works!*)
- Detailed distribution of the internal loadings is *unknown* so far [Fig. (b)]
- Resultant loadings (*resultant effects of the distribution of loading*) [Fig. (c)],  $F_R$  and  $M_R$ , can be *known* by applying equations of equilibrium
- The section or “cut” through plane can be arbitrary, and the resultant loadings will also change accordingly

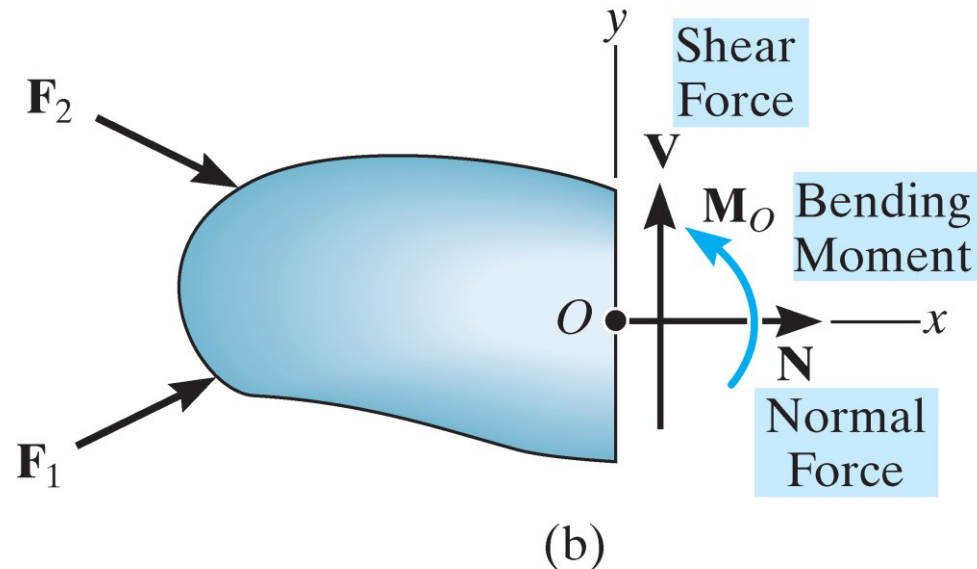
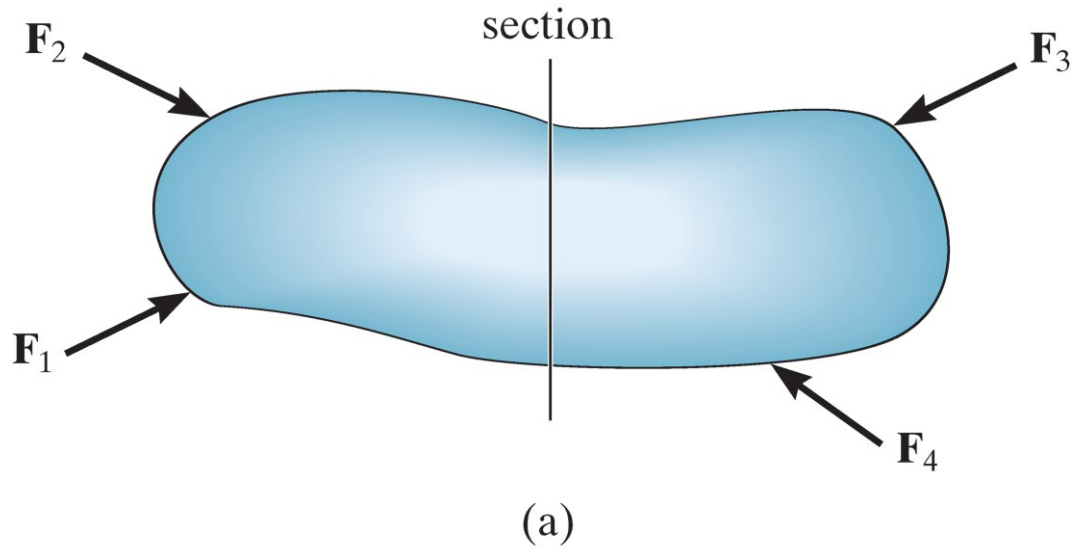
# EQUATION OF EQUILIBRIUM

- Types of resultant loadings (3D)

- **Normal force,  $N$ :** perpendicular to the section to prevent push or pull
- **Shear force,  $V$ :** in the section plane (relative movement of two segments)
- **Torsional moment or torque,  $T$ :** **twist** loads about an axis perpendicular to the section
- **Bending moment,  $M$ :** body **bend** about an axis lying within the plane of the section



# COPLANNAR LOADINGS (*xy* plane)



- In case of coplanar system of forces, only **N**, **V** and **M** components exist (**WHY?**)

✓ Determine **N**:

$$\sum F_x = 0$$

✓ Determine **V**:

$$\sum F_y = 0$$

✓ Determine **M**:

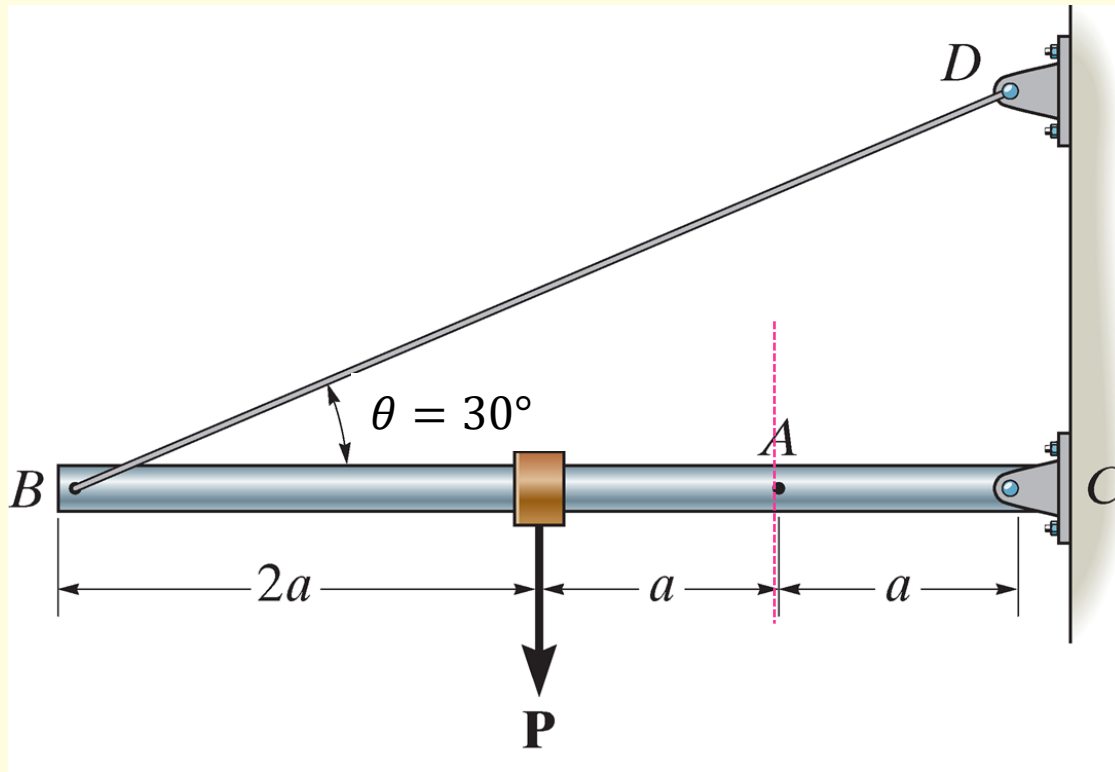
$$\sum M_z = 0$$



# ANALYSIS PROCEDURE OF RESULTANT INTERNAL LOADINGS

- **Step 1: Support Reactions**
  - Determine the reactions acting on the sectioned segment if there are any or relevant (*sometimes the free-body diagram of the entire body should be considered*)
- **Step 2: Free-Body Diagram**
  - Draw free-body diagram for one of the sectioned segment and indicate the unknown resultants (**N**, **V**, **M**, and **T**)
  - For the case of coplanar system of forces, only **N**, **V**, and **M** are applied
- **Step 3: Equations of Equilibrium (EoE)**
  - Determine unknown variables / quantities by applying EoE.

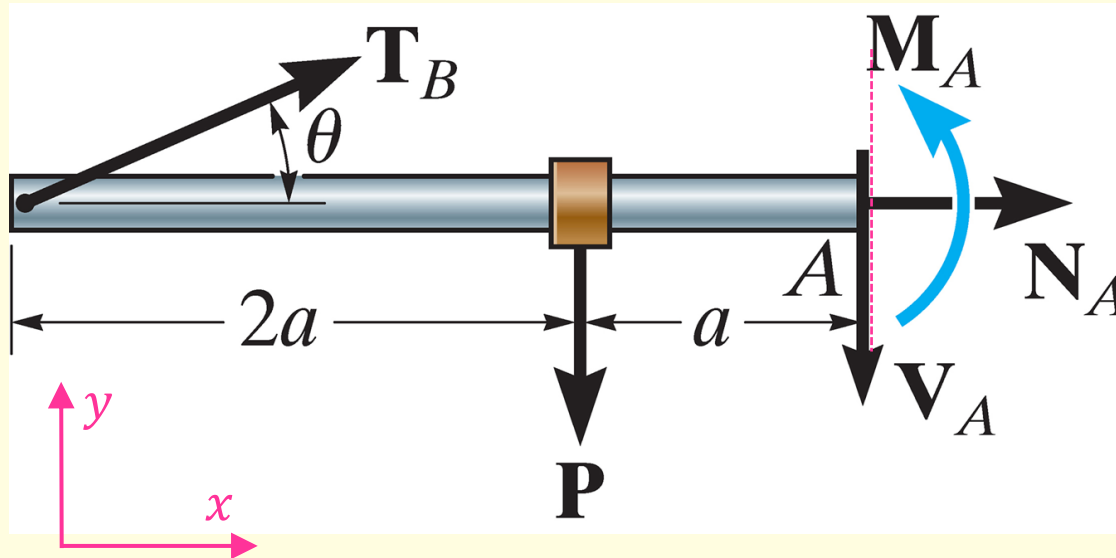
# IN-CLASS EXAMPLE



Find the resultant internal loading acting on the cross-section at point  $A$ , when the external load  $P = 10 \text{ kN}$ . Take  $a = 1 \text{ m}$ .

- **Important Rule**: when selecting the segment part to do free-body diagram, consider the one with as simple external loads as possible! Try to avoid the support reactions.

# SOLUTIONS TO IN-CLASS EXAMPLE

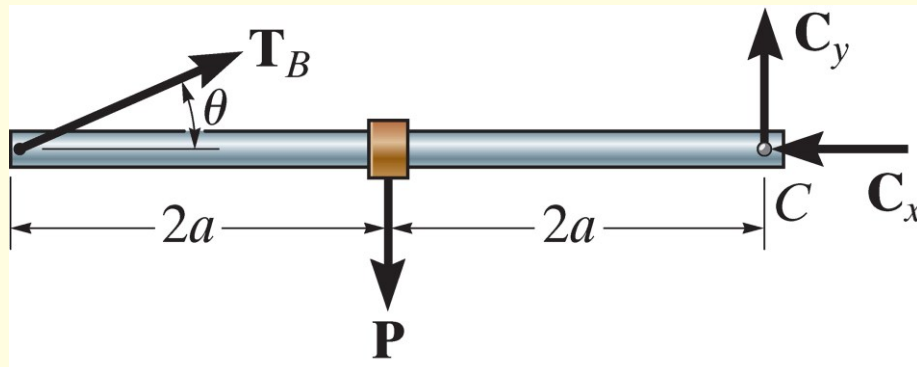


*For segment cutting through A (dashed line):*

$$\sum F_x = 0 \quad \rightarrow \quad N_A + T_B \cos \theta = 0 \quad (1)$$

$$\sum F_y = 0 \quad \rightarrow \quad V_A + P - T_B \sin \theta = 0 \quad (2)$$

$$\sum M_A = 0 \quad \rightarrow \quad M_A + P \times a - T_B \times 3a \times \sin \theta = 0 \quad (3)$$



*For entire body:*

$$\sum M_C = 0 \rightarrow T_B \times 4a \times \sin \theta = P \times 2a \quad (4)$$

Solving Eqs. (1) – (4):

$$T_B = P = 10 \text{ kN}, \quad N_A = -8.66 \text{ kN},$$

$$V_A = -5 \text{ kN}, \quad M_A = 5 \text{ kN}\cdot\text{m}$$

*The negative sign means the direction for N and V should be opposite to our presumed one.*

# RELEVANT PRACTICE

- Assignment 1, problem #1 (go to Blackboard under the “Assignments” folder, **Assignment 1 due: 8:30 AM, Monday, Sept. 13<sup>th</sup> 2021**)
- Optional Extra Exercise 1, problem #1 (go to Blackboard under the “Assignments” folder, **Extra Exercise 1 due: 8:30 AM, Monday, Sept. 13<sup>th</sup> 2021**)
- *Submit your solutions directly onto Blackboard*

# OUT-OF-CLASS READING

- **Textbook, pp. 4 – 14 (in particular examples 1.1 – 1.4)**